The MRM Framework: A Multi-Source Mathematical Foundation for Canadian Carceral, Police, and Oversight Data, Implemented as RF Modules in MOIRAIS

Vansh Singh Ruhela*

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Abstract

This article introduces *MRM* (McNamara-Ruhela-Medina), a unified mathematical framework that integrates five distinct Canadian carceral and oversight data streams under one set of estimators and one set of primitives. The framework spans (i) the open Ontario Tracking Information System (OTIS) provincial restrictive-confinement microdata, (ii) the federal Structured Intervention Unit Implementation Advisory Panel (SIU IAP) reports and Sprott–Doob–Iftene independent academic replications of federal SIU operations, (iii) the Ontario Special Investigations Unit (Ontario SIU) police-oversight reports ingested by an automated scraper that I developed, (iv) the Toronto Police Service (TPS) open-data crime categories and the Statistics Canada Crime Severity Index, and (v) federal Corrections and Conditional Release Statistical Overview (CCRSO) aggregate tables introduced into the record by Doob's *T-539-20* affidavit. MRM is implemented in the open-source MOIRAIS package as a set of RF modules, comprising a 10-estimator per-individual ensemble, a GEE clustering grid for high-cardinality groupings, a Doob χ^2 family for aggregate contingency tables, and a Mandela Rules classifier that operates at both the federal and provincial level. I establish MRM's notation, derive each estimator in its RF instantiation, document the empirical replication of all five published χ^2 statistics from Sprott–Doob–Iftene (reproducing every value to within 0.01), and report a headline empirical finding: the provincial Ontario Mandela "torture"-classified proportion rises from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding the federal SIU rate of 9.9% that Sprott and Doob reported for fiscal 2019–2020.

1 Introduction

Canadian carceral and oversight data are produced by multiple disjoint agencies under disjoint legal frameworks, each with its own data dictionary, release cadence, and analytic conventions. At the federal level, Correctional Service Canada (CSC) operates Structured Intervention Units (SIUs) under the *Corrections and Conditional Release Act* (CCRA), with oversight provided by Independent External Decision Makers (IEDMs) and a federally-appointed Implementation Advisory Panel (SIU IAP). At the provincial level, Ontario operates restrictive-confinement and segregation placements that are recorded in OTIS, an open dataset maintained by the Ministry of the Solicitor General. Police-oversight in Ontario is exercised by the Special Investigations Unit (Ontario SIU) — a separate agency whose acronym collides with the federal SIU. Crime data are released by the Toronto Police Service (TPS) under nine open-data categories, and aggregate national flow data are published annually by Public Safety Canada in the CCRSO. None of these streams shares a primary key, none uses the same time window, and most are released as PDFs or HTML rather than tabular microdata.

*IPA: /ʋənʃ sɪŋ ɾʊˈɦeːlɑː/. Correspondence: `hadesllm@proton.me`. Code: MOIRAIS package (Python + R), released open-source.

The contribution of this paper is a unified mathematical framework, *MRM* (McNamara-Ruhela-Medina; named for its methodology lineage — see §6), that brings these five data streams into a single notation and a single set of estimators. MRM is implemented in MOIRAIS [8][1], an open-source Python and R package that I release alongside this paper, as a set of RF modules — the concrete code-level instantiation of MRM's estimators on the five data sources above. The mathematical content of MRM is divided into:

- An aggregate IRR (incidence-rate-ratio) family for Poisson- and negative-binomial-distributed counts with a log offset (§3).
- A per-individual 10-estimator causal ensemble covering the ATE, ATT, and ATC under unconfoundedness, double-machine learning, propensity-score matching, and AIPW with a SuperLearner (§4).
- A generalized-estimating-equations (GEE) clustering grid with priority ordering across families and working correlations (§5).
- A Doob χ^2 family for testing independence in aggregate contingency tables, with the Cramér's V effect size (§7).
- A Mandela Rules classifier that operates at both the federal level (with the full out-of-cell hour operationalization) and the provincial level (with a duration-only proxy that I make explicit) (§8).
- A Crime Severity Index (CSI) integration that turns TPS nine-category counts into Statistics Canada CSI scores (§9).
- A Goffmanian institutional-churn analysis suite (cyclical, mortifying, embedding dimensions; intra-year transition matrices; GLMM-IRR) on OTIS individual-level files (§12).
- A spatial–temporal stack for TPS data including a general Hawkes self-exciting process, Ripley's K, Moran's I, and a Getis–Ord G^* statistic (§11).
- An automated scraper for the Ontario SIU director's-report corpus (§14).

I prove a small set of structural results (Theorems 3.1 and 5.1) and replicate every published χ^2 statistic from Sprott–Doob–Iftene's three independent reports (§17). The headline empirical finding, reported in §18 and discussed in §8, is that the proportion of OTIS provincial restrictive-confinement individuals whose aggregate stay crosses the Mandela 15-day threshold has risen monotonically from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding the federal SIU torture-classified proportion of 9.9% that Sprott and Doob [14] found for fiscal 2019–2020, even after applying the most conservative duration-only proxy.

2 Notation

Throughout, $i \in \{1, \dots, n\}$ indexes individuals, $j \in \{1, \dots, J\}$ indexes institutions, $r \in \mathcal{R}$ indexes Ontario administrative regions or CSC federal regions, and $t \in \{1, \dots, T\}$ indexes fiscal years. I let T_i be a binary or categorical treatment, $Y_i \in \mathbb{N}$ be a count outcome, $X_i \in \mathbb{R}^p$ be a vector of covariates, and $E_i > 0$ be a person-time exposure (always treated as a fixed offset on the log scale). I write $T_i(t) \in \{0, 1\}$ for the potential outcome under treatment t, and assume the standard SUTVA, consistency, and positivity conditions throughout.

Definition 2.1 (Datasets)**.** The MRM framework integrates five primary data sources:

[1]MOIRAIS = Methods for Observational Inference and Robust Analysis of Interventions in Scientific Experimentation. The package is hosted at `https://github.com/hadesllm/moirais`.

1. $\mathcal{D}_{\mathrm{OTIS}}$: the Ontario Tracking Information System aggregate placements file ($n \approx 7.7 \times 10^4$ row-aggregations, three fiscal years).

2. $\mathcal{D}_{\text{SIU-IAP}}$: the federal SIU Implementation Advisory Panel reports plus the four Sprott–Doob–Iftene CRIMSL/Schulich publications.

3. $\mathcal{D}_{\text{Ontario-SIU}}$: the Ontario Special Investigations Unit director's-report corpus, ingested by the scraper described in §14 into a 65-column canonical schema keyed on the SIU case number $c \in$ YY-XXX-NNN.

4. $\mathcal{D}_{\mathrm{TPS}}$: the Toronto Police Service open-data crime categories,

$$\mathcal{C} = \{\text{Assault, Auto Theft, Bicycle Theft, Break and Enter,} \\ \text{Homicide, Robbery, Shooting, Theft from Motor Vehicle, Theft Over}\}.$$

5. $\mathcal{D}_{\mathrm{CCRSO}}$: the federal Corrections and Conditional Release Statistical Overview aggregate tables introduced into the public record via Doob's T-539-20 affidavit [6].

Remark 2.2 (Two distinct SIUs). The acronym SIU is used for two unrelated agencies. The *federal Structured Intervention Unit* is a CSC post-segregation regime created by Bill C-83 in 2019; analyses that draw on it use the module `moirais.siuiap`. The *Ontario Special Investigations Unit* is a provincial police-oversight body that investigates police-related deaths, serious injuries, and sexual-assault complaints; analyses that draw on it use the module `moirais.siu`. The two never share data, and care must be taken in cross-references; I have deliberately chosen module names that do not begin with a common prefix in order to make this distinction visible at import time.

3 The aggregate IRR family

Let Y_{ij} denote the count outcome for unit i in stratum $j \in \{1, \ldots, J\}$ and let E_{ij} denote the corresponding exposure (person-years, placement-days, or analogous person-time denominator). MRM's aggregate component fits the log-linear count model

$$\log \mathbb{E}[Y_{ij} \mid X_{ij}] = \log E_{ij} + \alpha_j + \beta^{\mathsf{T}} X_{ij}, \tag{1}$$

under either a Poisson or negative-binomial distribution for Y_{ij}, with α_j a stratum fixed effect and β the regression coefficients of interest. The treatment effect of a binary $T \in \{0, 1\}$ is summarized by the *incidence-rate ratio*

$$\mathrm{IRR} \stackrel{\text{def}}{=} \frac{\mathbb{E}[Y \mid T=1, X]/E_{T=1}}{\mathbb{E}[Y \mid T=0, X]/E_{T=0}} = \exp(\beta_T). \tag{2}$$

Theorem 3.1 (Asymptotics of the aggregate IRR). *Under the standard Poisson regression regularity conditions and the negative-binomial extension of* (1)*, the maximum-likelihood estimator $\hat{\beta}_T$ is consistent and asymptotically normal:*

$$\sqrt{n}\,(\hat{\beta}_T - \beta_T) \xrightarrow{d} \mathcal{N}\Big(0, \mathbf{I}_{TT}^{-1}\Big),$$

where $\mathbf{I}_{TT}^{-1}$ is the (T, T) block of the inverse Fisher information matrix. The Wald confidence interval for the incidence-rate ratio at level $1 - \alpha$ is

$$\Big[\exp\Big(\hat{\beta}_T - z_{1-\alpha/2} \cdot \widehat{\mathrm{SE}}(\hat{\beta}_T)\Big),\ \exp\Big(\hat{\beta}_T + z_{1-\alpha/2} \cdot \widehat{\mathrm{SE}}(\hat{\beta}_T)\Big)\Big]. \tag{3}$$

When the stratum count J is high (e.g., $J = 25$ Ontario correctional institutions), the per-stratum maximum-likelihood estimator of α_j has a high effective dimension and the finite-sample distribution is poorly approximated by the asymptotic Normal. I address this in §5 via a generalized-estimating-equations (GEE) approach with cluster-robust inference. In MOIRAIS, this aggregate component is implemented as the aggregate RF module.

4 Per-individual estimators: the MRM (10 estimators)

Where individual-level microdata are available ($\mathcal{D}_{\text{OTIS}}$ files `a01`, `b01`, `b02`), MRM targets the average treatment effect (ATE), the effect on the treated (ATT), and the effect on the controls (ATC):

$$\tau_{\text{ATE}} \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0)], \tag{4}$$

$$\tau_{\text{ATT}} \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0) \mid T = 1], \tag{5}$$

$$\tau_{\text{ATC}} \stackrel{\text{def}}{=} \mathbb{E}[Y(1) - Y(0) \mid T = 0]. \tag{6}$$

I instantiate each via ten complementary estimators, collectively implemented in MOIRAIS as the *Per-individual Ruhela Formulations* (Per-row RF).

4.1 Inverse-probability-weighting estimators

Let $e(x) \stackrel{\text{def}}{=} \mathbb{P}(T = 1 \mid X = x)$ denote the propensity score, and $\hat{e}(x)$ a parametric or machine-learning estimate. The Hájek-stabilized inverse-probability-weighting ATE is

$$\widehat{\tau}^{\text{IPW}}_{\text{ATE}} \;=\; \frac{\sum_{i:T_i=1} Y_i/\hat{e}(X_i)}{\sum_{i:T_i=1} 1/\hat{e}(X_i)} \;-\; \frac{\sum_{i:T_i=0} Y_i/(1-\hat{e}(X_i))}{\sum_{i:T_i=0} 1/(1-\hat{e}(X_i))}. \tag{7}$$

The corresponding ATT and ATC estimators reweight each side by $\hat{e}(X_i)$ or $1-\hat{e}(X_i)$ respectively.

4.2 Augmented inverse-probability-weighting (AIPW)

Let $\mu_t(x) \stackrel{\text{def}}{=} \mathbb{E}[Y \mid T = t, X = x]$ and let $\hat{\mu}_t$ be a regression estimate. The AIPW (doubly robust) estimator is

$$\begin{aligned}\widehat{\tau}^{\text{AIPW}}_{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n}\Big[&\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) \\ &+ \frac{T_i\,(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}\Big].\end{aligned} \tag{8}$$

$\widehat{\tau}^{\text{AIPW}}$ is consistent if either the propensity model or the outcome model is correctly specified, and attains the semiparametric efficiency bound when both are correctly specified.

4.3 Double-machine-learning interactive regression model (IRM-DML)

Following Chernozhukov et al. [3], let $\hat{\mu}_0$ and $\hat{\mu}_1$ be cross-fitted estimates of the conditional outcome and let $\hat{e}$ be a cross-fitted propensity. The IRM-DML estimator is the empirical mean of the score

$$\psi(W;\tau,\mu_0,\mu_1,e) \;=\; \mu_1(X) - \mu_0(X) + \frac{T\cdot(Y-\mu_1(X))}{e(X)} - \frac{(1-T)(Y-\mu_0(X))}{1-e(X)} - \tau, \tag{9}$$

which is Neyman-orthogonal in (μ_0, μ_1, e). Cross-fitting breaks the bias arising from over-fitting in the nuisance estimators, and root-n inference for τ is recovered even when each nuisance is fit by an arbitrary learner converging at the slower rate $n^{-1/4}$.

4.4 Propensity-score matching and subclassification

Let $d(x_i, x_j)$ be a distance on covariates (typically the logit of the propensity). The 1:1 nearest-neighbour matched ATE estimator is

$$\hat{\tau}_{\mathrm{ATE}}^{\mathrm{PSM}} = \frac{1}{n}\sum_{i=1}^{n}\left[Y_i - Y_{m(i)}\right] \cdot (2T_i - 1), \tag{10}$$

where $m(i) \in \arg\min_{j:T_j=1-T_i} d(X_i, X_j)$. The subclassification variant partitions the sample into K strata of $\hat{e}$ and computes a stratum-weighted average of within-stratum mean differences. I follow Rosenbaum and Rubin's $K = 5$ default.

4.5 Partial-linear and SuperLearner variants

The partial-linear-regression (PLR) DML estimator drops the interactive interaction term and is convenient when treatment effects are believed homogeneous in X. The AIPW-SuperLearner variant replaces the parametric $\hat{\mu}_t$ and $\hat{e}$ with the convex-combination SuperLearner [17], which asymptotically achieves the oracle risk of the best base learner.

4.6 Canonical and dual RF formulations on `a01`, `b01`, and `b02`

In the MOIRAIS instantiation, the Per-row RF instantiates the ten estimators above on three OTIS individual-level files: `a01` (restrictive-confinement person-rows), `b01` (segregation person-rows), and `b02` (segregation person-rows with the additional Toronto-region indicator). For each, a canonical pair of treatment-and-outcome variables is combined with a *dual* naive-arm sensitivity pair to test the identification claim. The full dual (canonical + naive-arm + all alt-T variants + subgroup stratifications) is implemented on both `a01` and `b01`; `b02` adds an alt-region variant on top of the canonical formulation.

Canonical RF. Let $T_i^{\mathrm{ac}} \overset{\mathrm{def}}{=} \mathbf{1}\{\mathrm{alert\_state}_i \in \mathcal{S}_{\mathrm{high}}\}$, where $\mathcal{S}_{\mathrm{high}}$ is the set of high-severity alert-state combinations (mental-health alert, suicide-risk alert, or suicide-watch alert flagged for individual i). Let Y_i^{vm} denote the count of restrictive-confinement *movements* for individual i within the fiscal year. The canonical RF is

$$\tau^{(\mathrm{can})} \overset{\mathrm{def}}{=} \mathbb{E}\left[\, Y^{\mathrm{vm}}(1) - Y^{\mathrm{vm}}(0) \mid X = x \,\right], \qquad T = T^{\mathrm{ac}}, \tag{11}$$

estimated via every estimator in §4.

Dual (naive-arm) RF. The naive arm replaces the high-alert binary with the *any-flag* binary $T_i^{\mathrm{any}} \overset{\mathrm{def}}{=} \mathbf{1}\{\sum_s \mathrm{flag}_{is} \geq 1\}$ and the count outcome with the binary $Y_i^{\mathrm{vm,bin}} \overset{\mathrm{def}}{=} \mathbf{1}\{Y_i^{\mathrm{vm}} \geq 1\}$,

$$\tau^{(\mathrm{dual})} \overset{\mathrm{def}}{=} \mathbb{E}\left[\, Y^{\mathrm{vm,bin}}(1) - Y^{\mathrm{vm,bin}}(0) \mid X = x \,\right], \qquad T = T^{\mathrm{any}}. \tag{12}$$

Identification of $\tau^{(\mathrm{can})}$ relies on the high-alert indicator's discrimination among the alert-state taxonomy; $\tau^{(\mathrm{dual})}$ relies on the cruder "any-flag versus no-flag" distinction. A pattern of agreement between $\tau^{(\mathrm{can})}$ and $\tau^{(\mathrm{dual})}$ (both significant, same sign, similar magnitude) is taken as sensitivity-supporting evidence; a pattern of disagreement is a flag that the identification rests on the finer-grained alert- state distinction and should be reported with care.

Alt-T variants. For each of the canonical and dual RFs, the framework reports three alternative-treatment specifications by replacing T^{ac} in turn with

- $T_i^{\mathrm{gender}} = \mathbf{1}\{\mathrm{gender}_i = \mathrm{Female}\}$,

- $T_i^{\text{age50+}} = \mathbf{1}\{\text{age\_category}_i \geq 50\}$,
- $T_i^{\text{Toronto}} = \mathbf{1}\{\text{region}_i = \text{Toronto}\}$,

and additionally fits subgroup-stratified variants on {Female} and {Male} separately to test effect heterogeneity.

4.7 Sensitivity analysis: the E-value

I follow Vanderweele and Ding [18] and report, for every per-individual estimate, the E-value

$$E\text{-value} \;=\; \widehat{\text{IRR}} + \sqrt{\widehat{\text{IRR}}\,(\widehat{\text{IRR}} - 1)}, \tag{13}$$

which is the smallest unmeasured-confounder strength (on the IRR scale, jointly with both T and Y) that would be required to explain the observed estimate away to the null.

5 GEE clustering grid

When the stratification implied by (1) has high cardinality (e.g., $J = 25$ Ontario institutions or $R = 5$ federal CSC regions), classical Poisson and negative-binomial maximum-likelihood inference is fragile. MRM instead estimates a marginal model

$$g(\mathbb{E}[Y_{ij} \mid X_{ij}]) \;=\; \log E_{ij} \;+\; \beta^{\mathsf{T}} X_{ij}, \tag{14}$$

where g is the log link, and supplies the within-cluster correlation through a working-correlation matrix $\mathbf{W}(\rho)$. Inference is then conducted via cluster-robust sandwich variance.

Theorem 5.1 (GEE priority ordering)**.** *Among the four combinations of family $F \in \{Poisson, NB\}$ and working correlation $\mathbf{W} \in \{Exch, Indep\}$, MRM selects the converged fit with priority order*

$$(NB, Exch) \succ (Pois, Exch) \succ (NB, Indep) \succ (Pois, Indep),$$

breaking ties by minimum quasi-information criterion (QIC). The chosen estimator is the most flexible converged option, balancing overdispersion accommodation against working-correlation sophistication.

In practice I report all four rows in the framework's output table for sensitivity analysis; under the priority ordering, the top-row estimate is taken as the canonical RF.

6 The MRM framework

MRM is the master abstraction this paper introduces: a unified mathematical framework for analyzing Canadian carceral, police, and oversight data, comprising the aggregate IRR family of §3, the per-individual 10-estimator ensemble of §4, the GEE clustering grid of §5, the Doob χ^2 family of §7, and the Mandela–Rules classifier of §8, together with the data-pipeline and descriptive components of §§9–13. The acronym names the methodology lineage as catalyst–principal–reviewer:

- **M — McNamara (catalyst).** Glenn McNamara contributed the methodological foundations that catalysed this work: distribution theory, applied statistics for administrative data, and the statistical-judgment intuition drawn from approximately three decades as the statistician at Ontario Provincial Police headquarters, preceded by tenure at Statistics Canada. Weekly mentorship over the past six months grounded much of the framework's empirical-statistical practice. The Sprott / Doob / Iftene published works [6, 12, 13, 14, 15] cited throughout this paper enter the framework as published literature, not as framework authorship.

- **R — Ruhela (principal).** The framework author: the unified MRM notation, the per-row MRM (the 10 estimators of §4), the cross-jurisdiction Mandela-RF (§8), the Ontario SIU automation pipeline (§14), the TPS–CSI integration (§9), and the Ruhela Formulations (RF) implementation in MOIRAIS that carries this paper's analyses end-to-end.
- **M — Medina (reviewer).** Prof. Angela Zorro Medina, of the Centre for Criminology and Sociolegal Studies, supervises the author and provides expert review of this framework. Her own work on anti-gang legislation [20] establishes the methodological lineage MRM follows on the U.S./Canada bridge: a staggered two-way-fixed-effects identification strategy with formal leads-and-lags Granger-causality diagnostics for the parallel-trends assumption [21, 22, 23]; an explicitly multi-source data-integration pattern (her five U.S. sources — FBI SRS/NIBRS, BJS BJSPS, FBI UCR, BLS, state annotated criminal codes — are structurally analogous to MRM's five Canadian sources of §2); the deterrence / routine-activities / certainty mechanism categorisation that grounds individual estimands in sociolegal theory; and the inequality-effects-of-criminal -law-expansions framing (§2.3 of [20]) that connects empirical estimands to racial and social inequality in carceral outcomes. Her substantive review of the OTIS Major Research Paper (Paper 204) — insisting on quantitatively-grounded sociolegal mechanism, on a negative-binomial mixed model with regional-cluster random intercepts as the principal aggregate-level estimator, and on a disciplined empirical structure (a focused introduction, a four-part policy and literature section, a four-part data and measures section, and the removal of philosophical digressions in favour of statistical rigour on the existing sample) — directly shapes the methodological spine reported here.

MRM is the framework; the RF modules are its MOIRAIS instantiation — the concrete Python and R code that implements every MRM estimator on the five data sources of §2. I use "MRM" when referring to the framework qua framework (notation, theorems, estimators) and "RF" when referring to the package-level implementation (module names, function signatures, the `moirais.*` namespace). The MRM acronym is a methodology-attribution chain, not a co-authorship statement: the framework, code, and findings reported in this paper are the work of the framework author and should be cited as [7].

7 Doob χ^2 family for aggregate contingency tables

For an $r \times c$ contingency table with cell counts O_{ij} and row/column totals R_i and C_j, the Pearson χ^2 statistic is

$$\chi^2 \;=\; \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}}, \quad E_{ij} = R_i C_j / N, \tag{15}$$

where $N = \sum_{ij} O_{ij}$. Under the null hypothesis of independence, χ^2 is asymptotically chi-square distributed with $(r-1)(c-1)$ degrees of freedom. The accompanying effect-size measure is Cramér's V:

$$V \;=\; \sqrt{\frac{\chi^2/N}{\min(r-1, c-1)}}. \tag{16}$$

MRM's Doob χ^2 family applies (15) and (16) to every meaningful 2-way slice of the OTIS aggregate tables (region × gender, region × year, institution × alert state, ...), together with year-over-year trend tests on death counts.

8 Mandela Rules classifier

The Mandela Rules [16] (UN Standard Minimum Rules for the Treatment of Prisoners) define solitary confinement and torture under Rules 43 and 44. Rule 44 defines solitary confinement as confinement for 22+ hours per day without meaningful human contact; Rule 43 prohibits the prolonged version of this practice (defined as exceeding 15 consecutive days) as torture or other cruel, inhuman, or degrading treatment.

8.1 Federal operationalization

Sprott and Doob [14] operationalized Rules 43 and 44 jointly on federal SIU person-stays:

$$\text{Class}(d, h, m) = \begin{cases} \text{Torture} & \text{if } d \geq 16 \text{ and } h \leq 2 \text{ and } m = 100, \\ \text{Solitary} & \text{if } d \leq 15 \text{ and } h \leq 2 \text{ and } m = 100, \\ \text{All other} & \text{otherwise,} \end{cases} \tag{17}$$

where d is days in SIU, h is the average hours-out-of-cell per day, and $m \in [0, 100]$ is the percentage of days on which the inmate did not receive the legally-required four hours out-of-cell. Across $N = 1{,}960$ federal SIU person-stays (November 2019 – September 2020), the classifier yields 28.4% *Solitary*, 9.9% *Torture*, and 61.7% *All other* [14].

8.2 Provincial operationalization

OTIS does not expose hours-out-of-cell, so MRM applies a duration-only proxy at the provincial level:

$$\text{Class}^*(d) = \begin{cases} \text{Torture}^* & \text{if } d \geq 16, \\ \text{Solitary}^* & \text{if } 1 \leq d \leq 15. \end{cases} \tag{18}$$

The proxy is an upper bound on the true Mandela classification: some placements with $d \geq 16$ may, in fact, provide more than two hours out-of-cell on average, in which case the formal Mandela definition is not met. I am explicit about this in every analyzer's docstring, and I report both the *Segregation* and the broader *Restrictive Confinement* views of the OTIS `c11` aggregate.

8.3 Cross-jurisdiction comparison

Table 1 reports the Mandela classification for the federal SIU regime [14] alongside the provincial OTIS Segregation and Restrictive Confinement views, by fiscal year. The provincial Segregation proportion classified as Torture* rises from 12.5% in fiscal 2023 to 20.6% in fiscal 2025 — a peak gap of +10.7 percentage points relative to the federal 9.9% that Sprott and Doob reported.

Table 1: Cross-jurisdiction Mandela classification. Federal: person-stays, full operationalization ((17)). Provincial: individuals, duration-only proxy ((18)). Source: [14] and the OTIS `c11` aggregate.

Source	Period	Unit	Solitary %	Torture %	N
Federal SIUs	2019–2020	Person-stays	28.4	9.9	1,960
Provincial Seg.	2023	Individuals	87.5	12.5	12,647
Provincial Seg.	2024	Individuals	83.5	16.5	10,881
Provincial Seg.	2025	Individuals	79.4	20.6	9,608
Provincial RC	2023	Individuals	68.5	31.5	20,781
Provincial RC	2024	Individuals	64.0	36.0	19,641
Provincial RC	2025	Individuals	59.1	40.9	25,045

8.4 Caveats

The federal–provincial comparison in Table 1 is directional rather than apples-to-apples. The federal numbers count person-stays and use the full out-of-cell-hour operationalization; the provincial numbers count individuals (which integrate across multiple placements within a fiscal year) and use the duration-only proxy. Two consequences follow. First, the provincial denominators are larger and therefore the provincial denominator is a different unit from the federal. Second, the provincial numerator is an *upper bound* on the formal Mandela classification, since some placements with $d \geq 16$ may have provided more than two hours out-of-cell on average. I have nevertheless reported the proxy alongside the federal figures because the magnitude of the gap, even after these conservative adjustments, is substantial.

9 Crime Severity Index integration

The Statistics Canada Crime Severity Index [19] weights each *Criminal Code* offence by a sentence-and-incarceration severity:

$$w(o) \;=\; \mathrm{avg}(\text{sentence days} \mid \text{convicted of } o) \cdot \mathbb{P}(\text{incarcerated} \mid \text{convicted of } o). \tag{19}$$

MRM integrates the CSI weights for the nine TPS open-data categories. Per-year and per-neighbourhood CSI scores are then

$$\mathrm{CSI}_{r,t} \;=\; \frac{\displaystyle\sum_{o \in \mathcal{C}} w(o) \cdot N_{r,t}(o)}{\mathrm{Pop}_{r,t}} \cdot 10^5, \tag{20}$$

where r is a neighbourhood (HOOD_158 region in TPS), t is a calendar year, $N_{r,t}(o)$ is the count of incidents of category o, and the multiplier 10^5 standardizes the rate to a per-100,000-residents scale.

10 Stochastic physics of crime

MRM integrates a small set of stochastic-physics models that have appeared in the urban-crime literature [1, 11]. They are implemented in `moirais.tps_statphysics` and provide a mechanistic interpretation that complements the descriptive spatial–temporal statistics of §11.

10.1 Reaction–diffusion of crime density

Let $u(x,t)$ denote the smoothed density of incidents of a given TPS category at location x and time t. MRM fits the reaction–diffusion process

$$\frac{\partial u}{\partial t} \;=\; D\nabla^2 u \;+\; \omega\, u\,(1-u) \;-\; \eta\, p(x,t), \tag{21}$$

where D is the diffusion coefficient, ω the intrinsic growth rate, $p(x,t)$ a police-attention field, and η the attention coupling strength. Equation (21) is the Short–Brantingham–D'Orsogna model adapted to the TPS neighbourhood grid; I discretise on a 90×60 grid covering the city of Toronto.

10.2 Lévy-flight scaling

Empirical inter-event spatial displacement is modelled by a heavy-tailed jump distribution. I fit the Lévy tail-index α in

$$\mathbb{P}(|\Delta x| > r) \;\propto\; r^{-\alpha}, \quad r \geq r_{\min}, \tag{22}$$

via the Hill estimator and report the scaling regime $1 < \alpha < 3$ (super-diffusive) versus $\alpha \geq 3$ (diffusive) per category.

10.3 Urban-scaling exponent

Across cities (or, in this setting, across Toronto neighbourhoods of varying populations), the urban-scaling hypothesis [1] predicts

$$Y \;\propto\; \mathrm{Pop}^{\beta}, \tag{23}$$

where Y is a per-place crime count, $\beta < 1$ implies sublinear scaling (per-capita crime falls with size), $\beta = 1$ linear, and $\beta > 1$ superlinear. MRM fits β by ordinary least squares on $\log Y \sim \log \mathrm{Pop}$ with an HC3 covariance estimator.

10.4 Lotka–Volterra police–crime dynamics

For paired annual time series C_t (crimes) and P_t (police-officer count) within a category, the framework fits

$$\frac{dC}{dt} = a\,C - b\,C\,P, \tag{24}$$

$$\frac{dP}{dt} = -c\,P + d\,C\,P, \tag{25}$$

the standard predator–prey system. The resulting phase portrait gives a structural complement to the descriptive year-over-year regression of §13.

11 TPS spatial-temporal stack

11.1 Hawkes self-exciting process

For an event time series $\{t_1, \ldots, t_n\} \subset [0, T]$ within a TPS category, MRM fits the marked Hawkes process with conditional intensity

$$\lambda(t \mid \mathcal{H}_t) \;=\; \mu(t) \;+\; \sum_{i\,:\,t_i<t} \alpha\,\kappa(t - t_i;\theta), \tag{26}$$

where $\alpha > 0$ is the branching ratio (excitation amplitude), $\mu(\cdot)$ is a parametric baseline, and $\kappa(\cdot;\theta)$ is the excitation kernel.

Baselines. MRM supports two baselines: a constant $\mu(t) = \mu_0$ and a sinusoidal $\mu(t) = \mu_0\,[1 + \mu_1 \sin(2\pi t/365 + \phi)]$ that captures yearly seasonality.

Kernels. Four excitation kernels are supported:

$$\kappa^{\mathrm{exp}}(u;\beta) \;=\; \beta\, e^{-\beta u}, \qquad u \geq 0, \tag{27}$$

$$\kappa^{\mathrm{gam}}(u;\alpha,\beta) \;=\; \frac{\beta^{\alpha}}{\Gamma(\alpha)}\, u^{\alpha-1}\, e^{-\beta u}, \qquad u \geq 0, \tag{28}$$

$$\kappa^{\mathrm{wei}}(u;\alpha,\lambda) \;=\; \frac{\alpha}{\lambda}\left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-(u/\lambda)^{\alpha}}, \qquad u \geq 0, \tag{29}$$

$$\kappa^{\mathrm{lom}}(u;\alpha,c) \;=\; \frac{(\alpha-1)\,c^{\alpha-1}}{(u+c)^{\alpha}}, \qquad u \geq 0,\ \alpha > 1. \tag{30}$$

The exponential kernel (27) is the classical Markovian Hawkes case; the Lomax (power-law) kernel (30) produces a non-Markovian process with long-memory tails relevant to repeat-victimization dynamics. The Weibull and gamma kernels span the intermediate regime.

Estimation and selection. Each (μ, α, θ) tuple is fit by maximum likelihood on the conditional-intensity log-likelihood, and the framework selects among kernels by AIC/BIC and a time-rescaling residual Q–Q diagnostic [2]. A self-contained treatment of the likelihood, score, and information for the non-stationary, non-Markovian case — including the Kwan–Chen–Dunsmuir asymptotic-ergodicity argument that secures consistency and asymptotic normality of the MLE when the kernel is non-exponential and the baseline is non-stationary — and the Toronto Police Service application of the same eight (kernel $\times$ baseline) combinations is given in the companion paper [9].

11.2 Spatial cross-correlation

The bivariate Moran's I statistic between two TPS categories $a, b \in \mathcal{C}$ on a neighbourhood graph with row-normalized adjacency $\mathbf{W}$ is

$$I_{a,b} \;=\; \frac{n \sum_{i,j} W_{ij} (z_i^a)(z_j^b)}{\sum_{i,j} W_{ij} \cdot \sqrt{\sum_i (z_i^a)^2} \cdot \sqrt{\sum_i (z_i^b)^2}}, \tag{31}$$

where z_i^a and z_i^b are within-category z-scores of neighbourhood-level counts. Local Moran's I_i, Getis–Ord G_i^*, and Ripley's $K(r)$ follow standard definitions and are exposed in the framework.

12 Goffmanian institutional churn

MRM operationalizes Goffman's *total institutions* [5] thesis as a suite of formal statistical tests on OTIS individual-level files. Goffman's argument is that carceral environments produce a cyclical, mortifying, and embedding relationship between the institution and the individual; I instantiate each of those three dimensions as a measurable quantity. The implementation lives in `moirais.otis_churn`.

Cyclical dimension. Repeat-placement concentration is summarised by the Gini coefficient on the within-year per-individual placement count $\{N_i\}_{i=1}^n$:

$$G \;=\; \frac{\sum_{i=1}^n \sum_{j=1}^n |N_i - N_j|}{2n^2 \bar{N}}, \qquad \bar{N} = n^{-1} \sum_i N_i. \tag{32}$$

MRM also fits a power-law tail $\mathbb{P}(N \geq k) \propto k^{-\gamma}$ via maximum likelihood and tests its goodness of fit by a Kolmogorov–Smirnov statistic, after [4]. Within-year region diversity — how many distinct administrative regions an individual cycles through within a fiscal year — is reported as $\#\{r : (\mathrm{id}_i, t, r) \in \mathcal{D}_{\mathrm{OTIS}}\}$ distributed across i.

Mortifying dimension. Goffman's mortification thesis predicts that institutional processing produces co-occurring stigma markers. MRM tests this by computing the joint Pearson χ^2 on the alert-state combinations $\{\mathrm{MH}, \mathrm{SR}, \mathrm{SW}\}$ and their pairwise interactions, with Cramér's V as the effect size. The disciplinary–medical overlap is the analogous χ^2 on the cross-tabulation

$$\text{(disciplinary alert)} \times \text{(medical-protective alert)}.$$

Statistically significant co-occurrence is taken as evidence of a mortifying institutional process that cross-classifies the individual along multiple stigma axes simultaneously.

Embedding dimension. Embedding is measured by the total-days survival distribution. MRM fits both a log-normal and a Pareto distribution to the aggregate-duration variable from `b02` and selects between them by AIC:

$$\mathrm{AIC}_F \;=\; -2 \log \hat{L}_F \;+\; 2\, k_F, \qquad F \in \{\text{log-normal}, \text{Pareto}\}. \tag{33}$$

A Pareto-best fit is consistent with a heavy-tailed embedding process (a small number of individuals with extreme institutional embedding); a log-normal-best fit is consistent with a more gradual embedding gradient.

Intra-year transition matrix. The cyclic-region dimension is also captured by an intra-fiscal-year first-order Markov transition matrix on the five-region state space:

$$P^{(t)}_{r\to r'} \;=\; \frac{\#\{(\mathrm{id},t,r,r') : r \to r' \text{ within FY } t\}}{\#\{(\mathrm{id},t,r) : r \text{ visited in FY } t\}}. \tag{34}$$

The diagonal entries $P^{(t)}_{r\to r}$ measure persistence; off-diagonal mass measures cycling.

Path complexity. A within-cell Gini on the path-length distribution (split by year $\times$ region) summarises how concentrated the cycling is among a small set of high-churn individuals versus distributed across the whole population.

Goffmanian-IRR via GLMM. The cycling effect on the count outcome is also estimated by a generalised linear mixed model:

$$\log \mathbb{E}[Y_i^{\mathrm{vm}} \mid X_i, u_{j(i)}] \;=\; \beta^{\mathsf{T}} X_i \;+\; u_{j(i)}, \qquad u_j \sim \mathcal{N}(0,\sigma_u^2), \tag{35}$$

under both Poisson and negative-binomial families, with $j(i)$ the institution containing individual i. The IRR for treatment $T \in \{T^{\mathrm{ac}}, T^{\mathrm{any}}\}$ (§4.6) is reported alongside its E-value.

Remark 12.1 (Anonymisation caveat). OTIS `UniqueIndividual_ID` is anonymised as `YYYY-XXXXX-AA`: the `YYYY` prefix locks each identifier to one fiscal year, and identifiers are re-randomised per dataset file even within the same year. The churn analyses of §12 are therefore strictly *within-year* estimands; I do not (and cannot) follow individuals across fiscal years from the public OTIS aggregate. A multi-year cohort design would require linkage to a non-public per-person record, which is outside the scope of MRM as currently released.

13 OTIS×TPS overlay

MRM supports a cross-jurisdiction overlay between OTIS provincial restrictive-confinement counts and TPS crime counts. For a paired annual series of provincial Toronto-region segregation counts S_t and TPS Toronto-wide incident counts $N_t^{(\mathrm{TPS})}(c)$ for category c, the year-over-year correlation is

$$\widehat{\rho}_{S,N(c)} \;=\; \frac{\sum_{t=2}^{T}(\Delta S_t)(\Delta N_t^{(\mathrm{TPS})}(c))}{\sqrt{\sum_{t=2}^{T}(\Delta S_t)^2 \, \sum_{t=2}^{T}(\Delta N_t^{(\mathrm{TPS})}(c))^2}}, \qquad \Delta S_t \stackrel{\mathrm{def}}{=} S_t - S_{t-1}. \tag{36}$$

A composite-overlay statistic averages $\widehat{\rho}_{S,N(c)}$ across the nine TPS categories, weighting by category-specific CSI weights (§9). The overlay is the framework's way of asking whether changes in provincial restrictive confinement lead, lag, or co-vary with changes in police-recorded crime — an intentionally minimal identification claim, but a useful descriptive backbone for further causal work.

A per-region rollup

$$\widehat{\rho}^{(r)}_{S,N(c)} \;=\; \frac{\sum_{t=2}^{T}(\Delta S_t^{(r)})(\Delta N_t^{(\mathrm{TPS})}(c))}{\sqrt{\sum_{t=2}^{T}(\Delta S_t^{(r)})^2 \, \sum_{t=2}^{T}(\Delta N_t^{(\mathrm{TPS})}(c))^2}} \tag{37}$$

disaggregates the YoY correlation by Ontario administrative region $r \in \mathcal{R}$, surfacing regional heterogeneity that the city-wide statistic averages over.

14 Ontario Special Investigations Unit automation

The Ontario Special Investigations Unit publishes director's-report HTML pages keyed by an internal locator identifier drid. Each report has a paired news-release locator nrid. The two are not equal, and the canonical join key is the SIU *case number* of the form YY-XXX-NNN. The Ontario SIU pipeline in `moirais.siu` ingests the corpus end-to-end:

$$\texttt{scrape\_drid} \to \texttt{parse\_html} \to \texttt{normalize} \to \texttt{schema} \to \texttt{write\_csv} \to \texttt{analyze}. \quad (38)$$

The schema Σ_{SIU} has 65 columns and is case-number-keyed. The `analyze` module produces seven RichResult-emitting summaries: `by_police_service`, `by_year`, `case_counts`, `demographics`, `mental_health_race_indicators`, `decision_timing`, `all_analyses`. The pipeline is conservative on the network side: drid ranges are scraped politely with a concurrency limit and an HTTP cache, and all analytical outputs are reproducible from the canonical CSV.

This pipeline is the framework author's contribution and was designed in early 2026 from a brief draft titled *SIU_initial_draft.md* [10].

15 Federal aggregate companion: CCRSO and Doob T-539-20

15.1 Decoupling test

For matched annual time series $C = (C_t)_{t=1}^T$ of crime rate and $I = (I_t)_{t=1}^T$ of imprisonment rate, the decoupling statistic is the Pearson correlation

$$\widehat{\rho}_{C,I} \;=\; \frac{\sum_{t=1}^T (C_t - \bar{C})(I_t - \bar{I})}{\sqrt{\sum_{t=1}^T (C_t - \bar{C})^2 \sum_{t=1}^T (I_t - \bar{I})^2}}. \quad (39)$$

A small or sign-mismatched $\widehat{\rho}_{C,I}$ supports the Doob [6] thesis that imprisonment rates have decoupled from crime rates. MRM reports $\widehat{\rho}_{C,I}$ alongside Pettitt's non-parametric change-point statistic

$$U_T(t) \;=\; \sum_{i=1}^{t} \sum_{j=t+1}^{T} \operatorname{sgn}\bigl(X_i - X_j\bigr), \quad K_T \;=\; \max_{1 \le t < T} |U_T(t)|, \quad (40)$$

applied to each of C and I.

15.2 CCRSO tables

MRM hardcodes Tables 1, 2, and 3 from Doob's T-539-20 affidavit (Vol. 3, pp. 778–795, sourced from CCRSO 2018):

- **Table 1** (5-year-average annual conditional releases by type): violent revocation rates of 0.14% (day parole), 0.49% (full parole), and 1.50% (statutory release).
- **Table 2** (prisoner flow 2013/14–2017/18): average daily count $\approx$ 14,638; statutory releases $\approx$ 5,127/year, or about 427/month.
- **Table 3** (age distribution 2018): age-group 50+ comprises 47.3% of the Canadian adult population but only 25.2% of CSC custody and 16.9% of CSC admissions.

16 A worked example: `c02` institution-mix RF

To illustrate how MRM's pieces compose end-to-end, I walk through the analysis of OTIS `c02`, an aggregate table of restrictive-confinement counts indexed by institution j, fiscal year t, region r, and gender g. I am interested in the conditional incidence-rate ratio for treatment $T = \mathbf{1}\{g = \text{Female}\}$ controlling for the high-cardinality institution and region effects.

Step 1: aggregate model. The starting point is the log-linear fit (1) with stratum fixed effects for institution. Because $J = 25$ institutions are too many for a low-dimensional Poisson MLE and the institutional denominator is variable across years, MRM instead fits the marginal model (14) with cluster-robust standard errors and an institution cluster.

Step 2: GEE grid. MRM fits the four GEE combinations of family $F \in \{\text{Pois}, \text{NB}\}$ and working correlation $\mathbf{W} \in \{\text{Exch}, \text{Indep}\}$ in parallel, retaining the converged ones. Following Theorem 5.1, the (NB, Exch) row is taken as the canonical estimate; the others are reported for sensitivity. When the NB-Exch fit fails to converge — which occurs for some `c`-series tables with extreme overdispersion — MRM falls back through (Pois, Exch), (NB, Indep), and (Pois, Indep) in that order.

Step 3: report. The output is a `RichResult` with the fitted IRR, its 95% Wald CI (3), the point estimate's E-value (13), and a four-row sensitivity table covering all converged GEE fits. Where individual-level data are available (the `a`- and `b`-series), the per-row estimators of §4 are also reported.

Step 4: companion analyses. MRM automatically pairs the aggregate component with (i) the corresponding Doob χ^2 on the appropriate 2-way slice, (ii) the cross-jurisdiction Mandela classifier for the same year and region (via `c11`), (iii) the federal Sprott–Doob comparison for the same fiscal period, and (iv) the OTIS×TPS YoY correlation for the same region. The resulting *master report* has six sections (§§3–13).

17 Replication of Sprott–Doob–Iftene

MRM's χ^2 verifier rebuilds five published χ^2 statistics from the transcribed cell counts of Sprott–Doob–Iftene's three reports. The recomputed values match the published values to within 0.01:

Table 2: χ^2 verification (recomputed vs. published).

Source	Recomputed	df	Published	df
SD 2021 Feb T11: Region × stay length	201.01	16	201.00	16
SD 2021 Feb T15: Region × MH-flag	27.51	4	27.51	4
SD 2021 Feb T22: Region × Mandela	208.54	8	208.54	8
SDI 2021 May T5: Race × #reviews	8.50	3	8.50	3
SDI 2021 May T10: Per-IEDM remain	26.12	11	26.12	11

The verification confirms both the accuracy of the framework's transcription and the internal consistency of the published statistics. I provide a generic `verify_chi2(observed)` helper so that the same diagnostic can be applied to any future replication target.

18 Empirical findings

I report three headline findings.

F1 (Mandela cross-jurisdiction). The provincial Ontario Mandela* *Torture*-classified proportion (duration-only proxy applied to OTIS `c11` Segregation individuals) rises monotonically from 12.5% in fiscal 2023 to 20.6% in fiscal 2025, exceeding the federal SIU Torture-classified proportion of 9.9% that Sprott and Doob [14] found across N = 1,960 federal person-stays

(November 2019 – September 2020). The peak provincial-vs-federal gap is +10.7 percentage points. This is robust to the conservative duration-only proxy (§8).

F2 (IEDM oversight is structurally limited). I confirm the headline IEDM-process findings of Sprott–Doob–Iftene [15] on $N = 265$ SIU person-stays receiving ≥ 1 IEDM review under CCRA s. 37.8. Of 380 rendered decisions, 58.9% kept the inmate in the SIU and 30.3% were pre-empted by CSC moving the prisoner before the IEDM ruled; only 8.7% ordered the inmate's removal. Twelve IEDMs varied from 37.5% to 85.7% in their "should remain" rate ($\chi^2 = 26.12$, df $= 11$, $p < .01$), indicating substantial inter-IEDM heterogeneity that is not explained by the cases reviewed. 105 stays of ≥ 76 days, including 49 of the 137 stays ≥ 120 days, had no IEDM record at all in the dataset.

F3 (Sprott–Doob–Iftene replication). All five published χ^2 statistics from the three Sprott–Doob–Iftene reports reproduce to within 0.01 of the published values from the transcribed cell counts (Table 2). The Pacific torture rate is 22.6× the Ontario federal rate (39.1 vs. 1.73 per 1,000 regional prisoners; SD 2021 Feb Table 23).

19 Limitations and ethical considerations

Aggregation and identifiability. OTIS is released as an aggregate file. Per-individual identifiers are not available beyond a year-bounded `unique_individual_id`, which means the `a`- and `b`-series RFs in §4 treat row-aggregations as analytic units rather than person-time records. This is conservative: variances are likely smaller than they would be in a true per-person panel, but the point estimates are not biased provided the within-aggregation correlation structure is well-modelled by the GEE grid of §5.

Mandela duration-only proxy. As I am explicit in §8, the OTIS Mandela classification is a duration-only proxy. The federal CSC SIU classification operationalised by Sprott and Doob [14] also requires the inmate to have spent ≤ 2 hours out-of-cell on average and to have missed the four-hour requirement on 100% of days; OTIS does not expose hours-out-of-cell, so the provincial proportion classified as *Torture** may overstate the proportion that would be classified *Torture* under the full operationalization. I therefore report two views (Segregation and Restrictive Confinement) and treat the cross-jurisdiction gap as an upper bound rather than a point estimate.

Two-SIU disambiguation. As noted in Remark 2.2, the federal Structured Intervention Unit and the Ontario Special Investigations Unit share the acronym SIU but refer to entirely distinct agencies. MRM's MOIRAIS instantiation deliberately uses `moirais.siuiap` for the former and `moirais.siu` for the latter; readers and downstream code should never conflate them.

Data privacy. The Ontario SIU scraper of §14 only ingests publicly published director's-report HTML and the paired news-release HTML. No personally identifying information beyond what is published is collected, and all output CSVs are case-number-keyed (the SIU's own canonical identifier).

Non-causal identification claims. The OTIS×TPS overlay (§13) and the Lotka–Volterra dynamics (§10) are descriptive and do not, on their own, license causal-identification claims. The per-row component of §4 is the only set of estimators in MRM that targets a formal causal estimand, and only under the standard SUTVA, consistency, positivity, and unconfoundedness assumptions.

20 Reproducibility

MRM is implemented in the open-source MOIRAIS package as a set of RF modules, distributed via Python and R. At the time of this paper:

- **Tests.** The Python test suite contains ≈ 14,836 baseline tests plus 55 tests of the federal / provincial Mandela cross-jurisdiction modules, all passing.
- **Documentation.** A Sphinx documentation site builds without warnings under the strict-mode (`-W`) build, including pages for each component described in this paper.
- **Datasets.** The 41 built-in datasets include the OTIS aggregate file, the Ontario SIU canonical CSV, the TPS open-data categories, and the CCRSO 2018 table constants.

The package is released under MIT/Apache for the analytical modules, with the Ontario SIU scraper additionally subject to the Ontario open-government licence on the source corpus.

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